HW3

$\boxed{1.8}$ $P(V-Nb) = NkT$

a) Find Maxwell ~~Equa~~ relation involving $\left.\frac{\partial S}{\partial V}\right|_{T,N}$

Helmholtz Free Energy

$$F = E - TS \qquad dF = dE - TdS - SdT$$

$$= TdS - PdV - TdS - SdT$$

$$= -PdV - SdT$$

so $\left.\frac{\partial F}{\partial V}\right|_{T,N} = -P \qquad \left.\frac{\partial F}{\partial T}\right|_{V,N} = -S$

Cross derivative terms $\frac{\partial^2 F}{\partial T \partial V} = \left(\frac{\partial}{\partial T}\left.\frac{\partial F}{\partial V}\right|_{T,N}\right)_{V,N} = -\left.\frac{\partial P}{\partial T}\right|_{V,N}$

$$\frac{\partial^2 F}{\partial V \partial T} = \left(\frac{\partial}{\partial V}\left.\frac{\partial F}{\partial T}\right|_{V,N}\right)_{T,N} = -\left.\frac{\partial S}{\partial V}\right|_{T,N}$$

so $\left.\frac{\partial S}{\partial V}\right|_{T,N} = \left.\frac{\partial P}{\partial T}\right|_{V,N}$

b) Calculate $dE(T,V)$ show E is function of T only

$dE = TdS - PdV \qquad E = E(T,V)$

$$dE = T\frac{\partial S}{\partial T}dT + T\frac{\partial S}{\partial V}dV - PdV$$

Let's hold T constant so $dT = 0$

$dE = T\frac{\partial S}{\partial V}dV - PdV$ We just found $\frac{\partial S}{\partial V} = \frac{\partial P}{\partial T}$

so $\frac{\partial E}{\partial V} = T\frac{\partial P}{\partial T} - P \qquad P = \frac{NkT}{V-Nb} \qquad \frac{\partial P}{\partial T} = \frac{Nk}{V-Nb}$

so $\frac{\partial E}{\partial V} = T\frac{Nk}{V-Nb} - \frac{NkT}{V-Nb} = 0$ so E is function of T only

1.8 cont...

c) Show $\gamma = \frac{C_p}{C_v} = 1 + \frac{Nk}{C_v}$

$C_v = \left.\frac{\partial E}{\partial T}\right|_V$ $\quad C_p = \left.\frac{\partial E}{\partial T}\right|_P + P\left.\frac{\partial V}{\partial T}\right|_P$ $\quad$ Eq 1.14 of Kardar

$E = E(T)$ so $\left.\frac{\partial E}{\partial T}\right|_V = \left.\frac{\partial E}{\partial T}\right|_P = C_v$

$V = \frac{NkT}{P} + Nb \qquad \left.\frac{\partial V}{\partial T}\right|_P = \frac{Nk}{P}$

So $C_p = C_v + Nk$ and $\gamma = \frac{C_p}{C_v} = 1 + \frac{Nk}{C_v}$

d) $\frac{\partial E}{\partial T} = C_v \qquad dE = C_v dT$

What is dT? $\qquad T = \frac{P(V-Nb)}{Nk}$

since P & V are variable $\quad dT = \frac{(V-Nb)}{Nk}dP + \frac{P}{Nk}dV$

$dE = dQ + dW$ $\quad$ but $dQ = 0$ for adiabatic process

and $dW = -PdV$

So we have

$C_v dT = -PdV$ or $C_v\frac{(V-Nb)}{Nk}dP + \frac{C_v P}{Nk}dV + PdV = 0$

Divide by $P(V-Nb)$ $\qquad \frac{C_v}{Nk}\frac{dP}{P} + \frac{C_v}{Nk}\frac{dV}{(V-Nb)} + \frac{dV}{V-Nb} = 0$

$\frac{dP}{P} + \frac{dV}{V-Nb} + \frac{Nk}{C_v}\frac{dV}{V-Nb} = 0 \qquad \frac{dP}{P} + \gamma\frac{dV}{V-Nb} = 0 \qquad \frac{dP}{P} = -\gamma\frac{dV}{V-Nb}$

Integrate both sides $\int_{P_0}^{P}\frac{dP}{P} = \gamma\int_{V}^{V_0}\frac{dV}{V-Nb}$ $\quad$ I exchanged limits and got rid of minus sign

$\ln\frac{P}{P_0} = \gamma\ln\left(\frac{V_0-Nb}{V-Nb}\right) = \ln\left(\frac{(V_0-Nb)^\gamma}{(V-Nb)^\gamma}\right) \qquad P = P_0\frac{(V_0-Nb)^\gamma}{(V-Nb)^\gamma}$

So $P(V-Nb)^\gamma = P_0(V_0-Nb)^\gamma = \text{const}$